

General Relativity HW-6

PROBLEM 1. Consider a spherically symmetric 4-dimensional metric of the form

$$g_{\mu\nu}dx^\mu dx^\nu = -e^{2\alpha}dt^2 + e^{2\beta}dr^2 + r^2d\Omega^2$$

where α, β are functions of time t and radius r . Find all non-zero components of the Riemann tensor.

SOLUTION:

We firstly calculate the Christoffel symbol.

$$\Gamma_{jk}^i = \frac{g^{il}}{2}(\partial_j g_{lk} + \partial_k g_{jl} - \partial_l g_{jk})$$

Because g is diagonal, only the terms which have at least 2 identical indices will make contribution to Γ_{jk}^i .

$$\begin{aligned}\Gamma_{tt}^t &= \frac{-e^{-2\alpha}}{2}(-2\partial_t \alpha)e^{2\alpha} = \partial_t \alpha, \Gamma_{rr}^t = \frac{e^{-2\alpha}}{2}(2\partial_t \beta)e^{2\beta}, \Gamma_{rt}^t = \Gamma_{tr}^t = \frac{-e^{-2\alpha}}{2}(-2\partial_r \alpha)e^{2\alpha} \\ \Gamma_{rr}^r &= \partial_r \beta, \Gamma_{tt}^r = -\frac{e^{-2\beta}}{2}(2\partial_r \alpha)e^{2\alpha}, \Gamma_{\theta\theta}^r = -\frac{e^{-2\beta}}{2}(2r), \Gamma_{\varphi\varphi}^r = -\frac{e^{-2\beta}}{2}(2r \sin^2 \theta) \\ \Gamma_{\varphi\varphi}^\theta &= -\frac{1}{2r^2}2r^2 \sin \theta \cos \theta, \Gamma_{r\theta}^\theta = \Gamma_{\theta r}^\theta = \frac{1}{2r^2}2r = \frac{1}{r} \\ \Gamma_{r\varphi}^\varphi &= \Gamma_{\varphi r}^\varphi = \frac{1}{2r^2 \sin^2 \theta}2r \sin^2 \theta = \frac{1}{r}, \Gamma_{\varphi\theta}^\varphi = \Gamma_{\theta\varphi}^\varphi = \frac{1}{2r^2 \sin^2 \theta}2r^2 \sin \theta \cos \theta = \cot \theta\end{aligned}$$

And by the Christoffel symbol representation of Riemann tensor:

$$R_{jkl}^i = \partial_k \Gamma_{jl}^i - \partial_l \Gamma_{jk}^i + \Gamma_{k\lambda}^i \Gamma_{lj}^\lambda - \Gamma_{l\lambda}^i \Gamma_{kj}^\lambda$$

We have known that the Riemann tensor has in total $\frac{n^2(n^2-1)}{12}|_{n=4} = 20$ independent components. Let's list all the independent index sequences:

$$\begin{aligned}trtr, t\theta t\theta, t\varphi t\varphi, r\theta r\theta, r\varphi r\varphi, \theta\varphi\theta\varphi, trt\theta, trt\varphi, trr\theta, trr\varphi, tr\theta\varphi, t\theta t\varphi, t\theta r\theta, t\theta r\varphi, t\theta\theta\varphi, t\varphi\theta\varphi, t\varphi r\varphi \\ r\theta r\varphi, r\theta\theta\varphi, r\varphi\theta\varphi\end{aligned}$$

By calculating the Riemann tensor:

$$R_{rtr}^t = \partial_t \Gamma_{rr}^t - \partial_r \Gamma_{rt}^t + \Gamma_{t\lambda}^t \Gamma_{rr}^\lambda - \Gamma_{r\lambda}^t \Gamma_{tr}^\lambda =$$

$$\begin{aligned}
& \partial_t^2 \beta e^{2(\beta-\alpha)} + 2\partial_t \beta (\partial_t \beta - \partial_t \alpha) e^{2(\beta-\alpha)} - \partial_r^2 \alpha + \partial_t \alpha \partial_t \beta e^{2(\beta-\alpha)} + \partial_r \alpha \partial_r \beta - (\partial_r \alpha)^2 \\
& R^t_{\theta t \theta} = \partial_t \Gamma^t_{\theta \theta} - \partial_\theta \Gamma^t_{\theta t} + \Gamma^t_{t \lambda} \Gamma^\lambda_{\theta \theta} - \Gamma^t_{\theta \lambda} \Gamma^\lambda_{t \theta} = -(\partial_r \alpha) r e^{-2\beta} \\
& R^t_{\varphi t \varphi} = \partial_t \Gamma^t_{\varphi \varphi} - \partial_\varphi \Gamma^t_{\varphi t} + \Gamma^t_{t \lambda} \Gamma^\lambda_{\varphi \varphi} - \Gamma^t_{\varphi \lambda} \Gamma^\lambda_{t \varphi} = -(\partial_r \alpha) r e^{-2\beta} \sin^2 \theta \\
& R^r_{\theta r \theta} = \partial_r \Gamma^r_{\theta \theta} - \partial_\theta \Gamma^r_{\theta r} + \Gamma^r_{r \lambda} \Gamma^\lambda_{\theta \theta} - \Gamma^r_{\theta \lambda} \Gamma^\lambda_{r \theta} = -e^{-2\beta} + 2\partial_r \beta r e^{-2\beta} - \partial_r \beta r e^{-2\beta} + e^{-2\beta} = \partial_r \beta r e^{-2\beta} \\
& R^r_{\varphi r \varphi} = \partial_r \Gamma^r_{\varphi \varphi} - \partial_\varphi \Gamma^r_{\varphi r} + \Gamma^r_{r \lambda} \Gamma^\lambda_{\varphi \varphi} - \Gamma^r_{\varphi \lambda} \Gamma^\lambda_{r \varphi} = -e^{-2\beta} \sin^2 \theta + 2(\partial_r \beta) r e^{-2\beta} \sin^2 \theta - (\partial_r \beta) r e^{-2\beta} \sin^2 \theta + e^{-2\beta} \sin^2 \theta \\
& R^\theta_{\varphi \theta \varphi} = \partial_\theta \Gamma^\theta_{\varphi \varphi} - \partial_\varphi \Gamma^\theta_{\varphi \theta} + \Gamma^\theta_{\theta \lambda} \Gamma^\lambda_{\varphi \varphi} - \Gamma^\theta_{\varphi \lambda} \Gamma^\lambda_{\theta \varphi} = -\cos 2\theta - e^{-2\beta} + \cos^2 \theta
\end{aligned}$$

It's easy to verify that other possible free components are all zero (one way to see it is that Christoffel symbols need 2 identical indices to be non-zero, while the rest index sequence doesn't have pairs of indices). Thus all non-zero components are the symmetry and anti-symmetry form of the 6 non-zero independent components. The operations include switching the first two indices and the last two indices. Thus we see that there are in total $2 \times 2 \times 6 = 24$ non-zero components in Riemann tensor.

PROBLEM 2. Estimate a minimal mass of a Schwarzschild black hole.

SOLUTION:

We know the Schwarz radius:

$$r = \frac{2GM}{c^2} = \frac{8\pi G r^3}{3c^2} \rho \implies r = \sqrt{\frac{3c^2}{8\pi G \rho}}$$

We know that for neutrons, the maximum density is approximately

$$\rho_{max} \sim 10^{15} \text{ kg/m}^3$$

Thus the minimal mass is:

$$M_{min} = \frac{c^2}{2G} \sqrt{\frac{3c^2}{8\pi G \rho_{max}}} \sim 10^{32} \text{ kg}$$